



ENGINEERING MATHEMATICS-II

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FOURIER TRANSFORMS

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PROBLEMS:



1. Obtain the Fourier transform of $f(t)$ given by

$$f(t) = \begin{cases} 1 - t^2, & |t| \leq 1 \\ 0, & |t| > 1 \end{cases}$$

Solution: Fourier transform of $f(t)$ is given by

$$\mathcal{F}[f(t)] = F[\omega] = \int_{-\infty}^{\infty} f(t) \cdot e^{-i\omega t} dt$$

$f(t) = 1 - t^2$ for $-1 \leq t \leq 1$ and 0 otherwise.

PROBLEMS:



$$[f(t)] = F[\omega] = \int_{-1}^1 (1 - t^2) \cdot e^{-i\omega t} dt$$

$$\mathcal{F}[f(t)] = \mathcal{F} \left[(1 - t^2) \frac{e^{-i\omega t}}{-i\omega} - (-2t) \frac{e^{-i\omega t}}{(-i\omega)^2} + (-2) \frac{e^{-i\omega t}}{(-i\omega)^3} \right]_{-1}^1$$

$$\mathcal{F}[f(t)] = \left[-(1 - t^2) \frac{e^{-i\omega t}}{i\omega} - (2t) \frac{e^{-i\omega t}}{(\omega)^2} - (2) \frac{e^{-i\omega t}}{i\omega^3} \right]_{-1}^1$$

PROBLEMS:



$$\mathcal{F}[f(t)] = \left[0 - \frac{2}{\omega^2} (1 \cdot e^{-i\omega} - (-1) \cdot e^{i\omega}) - \frac{2}{i\omega^3} (e^{-i\omega} - e^{i\omega}) \right]$$

$$\mathcal{F}[f(t)] = \left[-\frac{2}{\omega^2} (e^{-i\omega} + e^{i\omega}) + \frac{2}{i\omega^3} (e^{i\omega} - e^{-i\omega}) \right]$$

$$\mathcal{F}[f(t)] = \left[-\frac{4}{\omega^2} \left(\frac{e^{i\omega} + e^{-i\omega}}{2} \right) + \frac{4}{\omega^3} \left(\frac{e^{i\omega} - e^{-i\omega}}{2i} \right) \right]$$

$$\begin{aligned} \mathcal{F}[f(t)] &= +\frac{4}{\omega^3} \sin \omega \quad \left(\because \cos x = \frac{e^{ix} + e^{-ix}}{2}, \sin x \right. \\ &= \left. \frac{e^{ix} - e^{-ix}}{2i} \right) \end{aligned}$$

$$\therefore \mathcal{F}[f(t)] = 4 \left[\frac{\sin \omega - \omega \cos \omega}{\omega^3} \right]$$

PROBLEMS:

2. Obtain the Fourier transform of $f(t)$ given by

$$f(t) = \begin{cases} 1, & -T \leq t \leq T \\ 0, & |t| > T \end{cases}$$



PROBLEMS:



Solution:

$$\mathcal{F}[f(t)] = F[\omega] = \int_{-T}^T 1 \cdot e^{-i\omega t} dt$$

$$\mathcal{F}[f(t)] = \left[\frac{e^{-i\omega t}}{-i\omega} \right]_{-T}^T$$

$$\begin{aligned}\mathcal{F}[f(t)] &= \frac{-1}{i\omega} (e^{-i\omega T} - e^{i\omega T}) \\ &= \frac{1}{i\omega} (e^{i\omega T} - e^{-i\omega T}) \\ &= \frac{2}{\omega} \cdot \frac{(e^{i\omega T} - e^{-i\omega T})}{2i}\end{aligned}$$

$$\therefore \mathcal{F}[f(t)] = \frac{2 \sin \omega T}{\omega}$$



THANK YOU

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